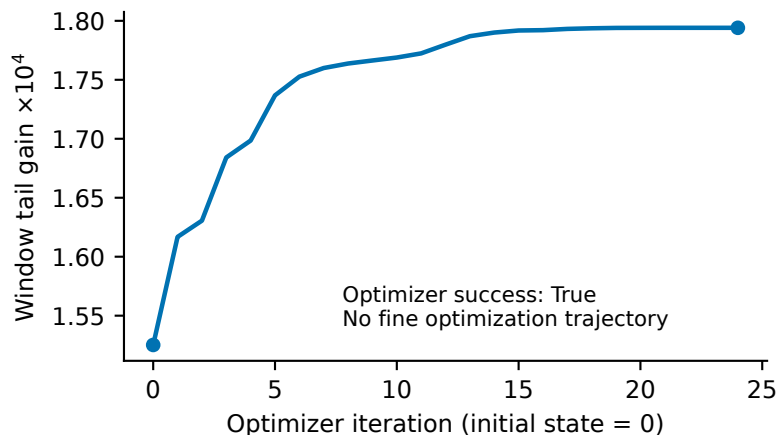
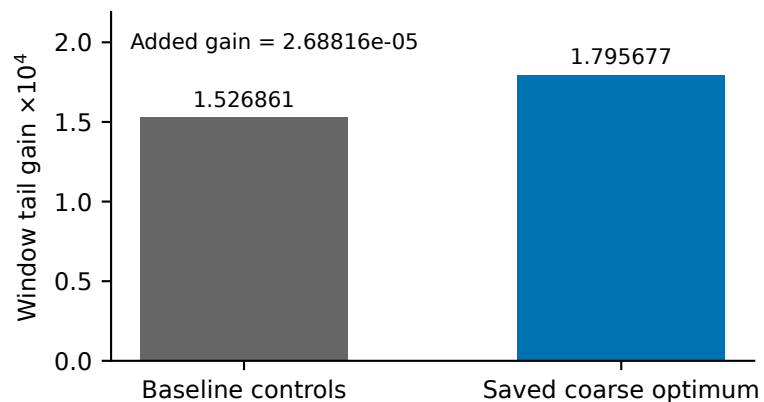


Constrained control of signed electron tail-energy gain

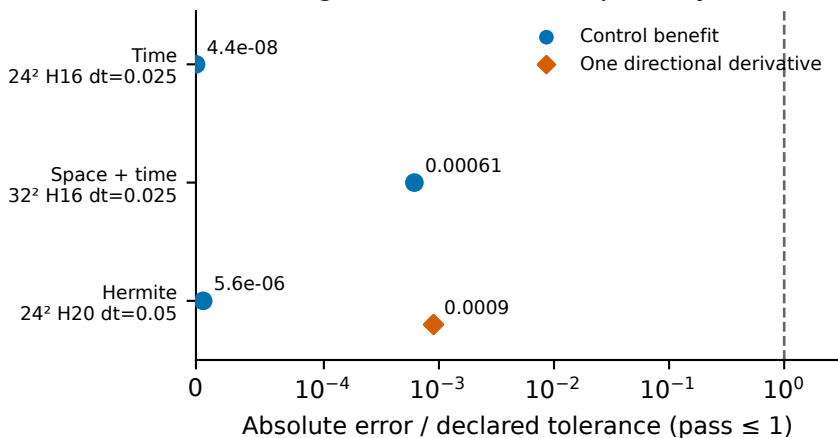
(a) Coarse training only
16² / H8 / dt=0.1



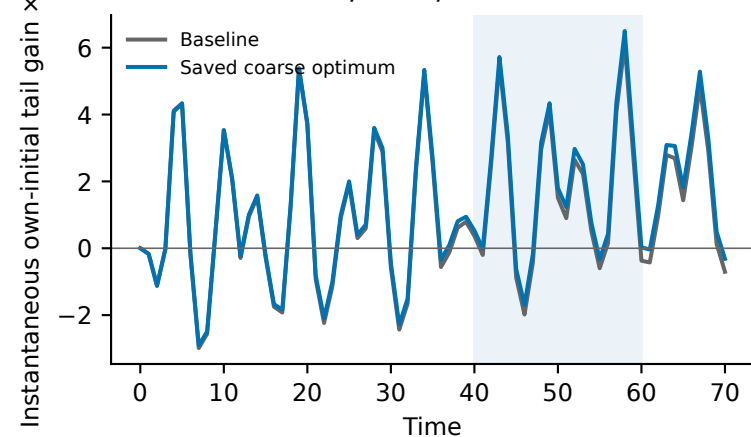
(b) Fine frozen-control reevaluation
24² / H16 / dt=0.05



(c) Agreement with fine primary case



(d) Frozen controls: sampled time history
24² / H16 / dt=0.05



Gain = weighted [40, 60] tail(Ct) – tail(C0), with a fixed initial thermal reference.

Tolerance = 1e-10 + 0.05|reference|. Refinement is reevaluation, not fine reoptimization.

Coarse endpoint local-negativity gate failed. Signed tail gain alone does not establish nonthermal acceleration.